MIL-S-48747B(AR)
14 October 1982
SUPERSEDING
MIL-S-48747A(AR)
22 August 1979

MILITARY SPECIFICATION

SAFE AND ARMING DEVICE FOR MINES, M718/M741 SYSTEM

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings, and provides quality assurance provisions for the assembly of a Safe and Arming Device for Mines used in the M718/M741 System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions General Specification For.
MIL-D-48120	-	Detonator, Electric: XM100; Parts for and Loading, Assembling and Packing.
MIL-D-63197	-	Detonator Delay Electric: PA506; Parts For, and Load, Assembling and Packing.
MIL-L-63246	-	Lead Assemblies for S&A Mechanism for Scatterable Mines; Parts for, and Loading, Assembling and Packing.

FSC 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-63383 - Safe and Arming (Non-Hazardous) Assembly For Mines M718/M741 System
MIL-P-63238 - Projectile, 155MM, AT, M718/M741 Loading, Assembling and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-453 - Inspection, Radiographic

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent herein. Code of Federal Regulations, Title 49 - Transportation, Parts 100-199 (The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision").

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9278015 - Safe and Arming Device for Mines, M718/M741 System

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicble drawings and spectifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on drawings listed in paragraph 2.1, and associated drawings, and with all requirements specified in applicable specifications and standards.

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3.3 Radiographic examination. The safe and arming device shall be examined for a fully non-armed condition in accordance with MIL-STD-453 and 4.5.4 and in accordance to (Dwg. 9278015). Further radiographic requirements shall include clear definition of all explosive components as well as flexible cable assembly and slider lock.

3.4 Environmental and safety tests.

3.4.1 Jolt. The safe and arming device shall conform with the requirements of MIL-STD-331, Test 101 (Jolt) when tested in accordance with 4.5.8.1.

3.4.2 Jumble. The safe and arming device shall conform with requirements of MIL-STD-331, Test 102 (Jumble) when tested in accordance with 4.5.8.2.

3.4.3 Transportation vibrations. The safe and arming device shall conform with the requirements of MIL-STD-331, Test No. 119 (Transportation Vibration) when tested in accordance with 4.5.8.3.

3.4.4 Static detonator safety. The safe and arming device shall conform with the requirements of MIL-STD-331, Test 115 (Static Detonator Safety) when tested in accordance with 4.5.8.4.

3.4.5 Ballistic functioning. The safe and arming device shall arm and all explosives shall function after being subjected to the following ballistic environments over the temperature range of -25°F to +145°F.

Charge zone 3 in the M109A1 weapon.
Charge zone 8 in the M109A1 weapon.

3.4.6 "G" weight lock spring position. A properly positioned "G" weight lock spring is not kinked, is contained within the lock recess and has its flat end entirely against the surface of the spring retainer.

3.5 Explosive functioning.

3.5.1 Main explosive train. The output of the main explosive leads (Dwg. 9277795) shall result and comply with the Table I Dent Test Requirements when tested in accordance with 4.5.6.1.

TABLE I - DENT TEST REQUIREMENTS

Depth of Depression (Inches Minimum)	Temperature (°C)
0.009	-40°C ± 3°C
0.011	+25°C ± 3°C
0.012	+60°C ± 3°C

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3.5.2 Aft ejection charge assembly (Dwg. 9277858). The aft ejection charge explosive elements shall function causing the propellant to ignite, resulting in a minimum of one hundred and sixty thousandths (0.160) inch deformation in the aluminum test plate at temperatures between minus forty degrees centigrade to plus sixty degrees centigrade.

3.6 Resistance measurements.

3.6.1 Detonator short circuit. The resistance of the short circuit across the detonators on the flexible cable shall be less than fifteen thousandth (0.015) ohms when tested in accordance with 4.5.5.2.

3.7 Data requirements. The contractor shall forward copies of the results of any of his examinations or tests to the designated address, as required by the contract (See 6.5).

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship - parts and assemblies. Parts and assemblies shall be devoid of burrs, chips, dirt, grease, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated. Exterior surface coatings shall be continuous, except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

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Part Description	Drawing	Quantity
Slider Assembly	9278011	10
Safe and Arming Subassembly	9278015	10
Safe and Arming Assembly	9278015	65
Chamber Assembly	9277986	10
Aft Ejection Charge Assembly	9277858	25
Setback Pin Assembly	9317617	10
Safe and Arming (Non-Hazardous) Assembly for Mines M718/M741 System	9317620	108

4.3.2 Inspection to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

The first article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or tests specified in this specification, MIL-S063383. or any of the requirements of the applicable drawings. In addition, the first article sample shall be rejected if a total of four (4) or more safe and arming assemblies fail to arm and/or function the explosive elements after being subjected to the ballistic tests. or two (2) or more assemblies fail to arm and/or function the explosive elements after ballistic testing in any one (1) temperature test group.

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TABLE II. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Safe and Arming Device for M718/M741 System				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Slider Assembly (Dwg. No. 9278011)				
	Examination for defects	10		3.2	4.4.2.1
	Safe and Arming Subassembly (Dwg. No. 9278015)				
	Examination for defects	10		3.2	4.4.2.2
	Non-Arming Test	10		3.2	4.5.1
	Non-Arming Test of Centrigugal Weights	10		3.2	4.5.3
	Arming Test	10		3.2	4.5.2
	Safe and Arming Assembly (Dwg. No. 9278015)				
	Examination for defects	10		3.2	4.4.2.4
	Tests				
	Jolt	5		3.4.1	4.5.8.1
	Jumble	5		3.4.2	4.5.8.2
	Transportation Vibration	5		3.4.3	4.5.8.3
	Static Detonator Safety	15(1)		3.4.4	4.5.8.4
	Explosive Tests	25(2)		3.5.1	4.5.6.1
	Detonator Resistance	25		3.2	4.5.5.1
	Detonator Short Circuit	25		3.5.3	4.5.5.2
	Closure, Firing Pin Stake Push Off Test	5		3.2	4.5.7
	Closure, Slider Stake Push Off Test	5		3.2	4.5.7
	Radiographic Examination	10		3.3	4.5.4

NOTES:

(1) Five (5) each assemblies shall be tested at the temperature specified in MIL-STD-331.

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TABLE II. First Article Inspection - continued

CLASSIFICATION OF DEFECTS & TESTS MIL-S-48747B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Safe and Arming Device for M718/M741 System				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Chamber Assembly (Dwg. No. 9277986)				
	Examination for defects	10		3.2	4.4.2.5
	Aft Ejection Charge Assembly (Dwg. No. 9277858)				
	Examination for defects	25		3.2	4.4.2.6
	Functioning test	25		3.5.2	4.5.6.2
	Setback Pin Assembly (Dwg. No. 9317617)				
	Examination for defects	10		3.2	4.4.2.7
	Safe and Arming (Non-Hazardous) Assembly for Mines, M718/M741 System (Dwg. No. 9317620)				
	Examination for defects	108		3.2	MIL-S-63383
	Ballistic tests (Note 3)				
	Zone 3 -25°F	36(4 Rounds)		3.4.5	4.5.11
	Zone 8 70°F	36(4 Rounds)		3.4.5	4.5.11
	Zone 8 +145°F	36(4 Rounds)		3.4.5	4.5.11

NOTES: (2) 15 Ambient, 5 Hot, 5 Cold.
(3) Configured per Dwg. 9317620.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of safe and arming assemblies & sub-assemblies shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Explosive detonators, leads, and MDF from the same lot interfix number from one manufacturer.

c. Propellant from the same lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278011
4.4.2.1	Slider Assembly				NEXT HIGHER ASSEMBLY 9278015
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Center charge lead assembly missing or assembled incorrectly		100%	3.2	Visual
102	Transfer charge lead assembly missing, or assembled incorrectly		100%	3.2	Visual
103	Groove pin missing, or assembled incorrectly		100%	3.2	Visual
104	Lead holder transfer charge stake improper		100%	3.2	Visual
105	Lead holder missing		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Safe and Arming Subassembly		9278015(3)
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(5)				
1	Spring, centrifugal lock missing		100%(1)	3.2	Visual/Manual
2	Holder, spring, missing, improperly oriented or loose		100%	3.2	Visual/Manual
3	Lock, centrifugal missing		100%	3.2	Visual
4	Any spring missing or improperly positioned		100%	3.2	Visual/Manual
5	Presence of "G" weight		100%	3.2	Visual/Manual
6	Presence of ball		100%	3.2	Visual/Manual
7	Presence of "G" weight		100%	3.2	Visual/Manual
8	Presence of set back pin assembly		100%	3.2	Visual/Manual
9	Non-arming test		100%	3.2	4.5.1
10	Non-arming centrifugal lock test		100%(4)	3.2	4.5.3
11	Deformation of centrifugal lock hole(s) after spring holder staking	4.4.3.4		3.2	4.5.10
Major					
101	Presence of more than one spring per cavity		100%	3.2	Visual/Manual
102	Staking missing or improper		100%	3.2	Visual
103	Presence of rear locks		100%	3.2	Visual
104	Presence of rear lock springs		100%	3.2	Visual
105	Any closure missing, loose or improperly staked		100%	3.2	Visual/Manual
106	Arming test		100% (2)(4)	3.2	4.5.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Safe and Arming Subassembly				9278015(3)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor 201	Evidence of poor workmanship		100%	3.9	Visual

Notes:
(1) Manual inspection is defined as the allowable probing of the unit by means of an approved instrument which will not cause damage to the sample.
(2) Test is to be performed prior to inserting rear lock assembly.
(3) Parts list shall be in accordance with 4.5.1.

NOTES:
(4) Arming test shall always precede the non-arming tests.
(5) Para. 4.5.12 applies whenever critical defects are discovered.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9278015
4.4.2.4	Safe and Arming Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical (2)					
1	Assembly in partially armed or armed state or missing components (radiographic)		100%	3.3	4.5.4
2	Jolt	5/0/1		3.4.1	4.4.3/4.5.8.1
3	Jumble	5/0/1		3.4.2	4.4.3.3/4.5.8.2
4	Static detonator safety	5/0/1		3.4.4	4.4.3.3/4.5.8.4
5	"G" weight lock spring missing or improperly positioned		100%	3.4.6	Visual
Major					
101	Interrupter free to move		100%	3.2	Note 4 of Dwg. No. 9278015
102	Chamber assembly improperly staked		100%	3.2	Visual
103	Grounding clip missing or improperly inserted		100%	3.2	Visual
104	Detonator(s) not fully seated		100%	3.2	Visual/Manual
105	Solder improper		100%	3.2	Visual
106	Presence of all closure discs		100%	3.2	Visual/Manual
107	All main charge lead assemblies correctly inserted and staked		100%	3.2	Visual
108	Shield assembly missing, loose, not staked, or improperly installed		100%	3.2	Visual/Manual
109	Detonator resistance		100%	3.2	4.5.5.1
110	Explosive test			3.5.1	4.4.3.1.1/4.5.6.1
111	MDF damage		100%	3.2	Visual
112	Detonator not shorted		100% (1)	3.6	4.5.5.2

NOTES: (1) Verify short circuit after resistance measurements have been taken and the short circuit has been replaced.
(2) Para. 4.5.12 applies whenever critical defects are discovered.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Safe and Arming Assembly				9278015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Cable spacer missing or improperly installed		100% (3)	3.2	Visual
114	Closure firing pin stake push off test			3.2	4.4.3.2/4.5.7
115	Closure, slider stake push off test			3.2	4.4.3.2/4.5.7
116	Transportation vibration	5/0/1		3.4.3	4.4.3.3/4.5.8.3
117	Closure, slider oriented properly		.40%	3.2	Visual
118	Ballistic tests -				
	Zone 3 Ambient	18(2 Rounds)	(4)	3.4.5	4.5.11
	Zone 8 Ambient	18(2 Rounds)	(4)	3.4.5	4.5.11
119	Detonator placement incorrect		0.40%	3.2	Visual
120	Height of solder joint(s), max.		0.40%	3.2	Gage
121	Solder workmanship under flexible cable improper				4.4.3.5/4.5.13
Minor 201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES: (3) Inspection must be made during assembly operation.
(4) Test for information only - configuration per Dwg. 9317620.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Chamber Assembly				9277986
					NEXT HIGHER ASSEMBLY 9278015
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major					
101	Aft ejection charge assembly missing or not fully seated in dual MDF chamber		100%	3.2	Visual/Manual
102	Pin, locating missing or improperly inserted		0.40%	3.2	Visual
103	Stake missing or improper		0.40%	3.2	Visual
104	Molded insert(s) missing or improperly seated		0.40%	3.2	Visual
105	Crack or break in MDF		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48747B(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Aft Ejection Assembly	1	1	9277858
				NEXT HIGHER ASSEMBLY 9277986

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Functioning			3.5.2	4.4.3.1.2/4.5.6.2
Minor 201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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MIL-S-48747B(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.7	Setback Pin Assembly	1	1	9317617
				NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Protrusion of stop pin		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		1.0%	3.9	Visual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.3.1 Explosive testing.

4.4.3.1.1 In-line explosive functioning. One hundred and twenty five (125) samples shall be randomly selected from each lot and tested as follows at the specified temperatures:

Temperature	-40°C ±3°C	+25°C ±3°C	+60°C ±3°C
Quantity	20	85	20

The lot shall be rejected if the total of rejects from all temperatures exceeds the following requirements.

Criteria	Accept No.	Reject No.
Dent Measurement or Dud	1	2
Delay Time	1	2
Aft Ejection Charge Initation	1	2

4.4.3.1.2 Aft ejection charge assembly. One hundred and twenty-five (125) samples shall be randomly selected from each lot and tested as follows at the specified temperatures.

Temperature	-40°C ±3°C	+25°C ±3°C	+60°C ±3°C
Quantity	20	85	20

The lot shall be rejected if the total of rejects from all temperatures exceeds the following requirements:

Criteria	Accept No.	Reject No.
M5 Propellant Initiation or Dud	1	2
Dent Measurement	1	2
Blow Through	6	7

4.4.3.2 Stake push off test. A "Push off Test" shall be performed on the front and rear housing closures after being staked to the housing, to check for integrity of staking. The test shall be performed on two samples each, prior to the start of each shift, and then performed on one sample each, after every two hours production and at the end of the shift, until three consecutive shifts have been accepted. Thereafter, the test may be performed on one sample each, prior to the start of each shift, and one sample each after four hours production and at the end of the shift. Failure of a test sample at any time will cause the test to revert to the initial test conditions until three consecutive shifts have been accepted.

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4.4.3.3 Environmental and safety testing. Safe and arming samples shall be tested in accordance with 4.5.8. Acceptance of samples shall be in accordance with the provisions of each applicable test methods of MIL-STD-331. Any unit failing to comply with the requirements shall cause that environmental test group to be classified defective and the lot of S&A's shall be rejected.

4.4.3.4 Centrifugal lock hole inspection. The inside diameter of the centrifugal lock holes in the S&A housing shall be inspected after staking of the centrifugal lock spring holders. The inspection shall be performed on three (3) centrifugal lock/housing subassemblies from each staking tool at the start of each shift and then on one (1) subassembly randomly selected from each group of 100 produced continuously from each staking tool. The contractor shall not enter into production with any staking tool unless all three (3) samples pass the initial inspection. Both centrifugal lock holes on each sample shall be inspected. Inspection shall be performed in accordance with 4.5.10. The contractor may request a change in frequency of the inspection provided he presents substantive evidence to support the request.

4.4.3.5 Solder workmanship. The contractor shall assure that the solder process/procedure does not contaminate or create potential short circuit conditions on the bottom side of the flexible cable due to solder flow thru the holes. The contractor shall examine the interface between the detonators and the cable for flux, excessive solder or other unacceptable criteria specified in the solder standard listed on the drawing. The examination shall be performed, after removal from the housing, on five (5) detonator/cable assemblies prior to the start of each shift, after every two (2) hours, and at the end of the shift until three (3) consecutive shifts have been accepted. Thereafter, the examination may be performed on two (2) assemblies prior to the start of each shift and one sample after four (4) hours of production and at the end of the shift. Failure of a sample at any time will cause the examination to revert to the initial conditions until three (3) consecutive shifts have been accepted.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the tables beginning with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval, in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Non-Arming of safe and arming subassembly. The subassembly shall be tested for the specified non-arming requirements using the following parts as a minimum for the sample:

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Part No.	Nomenclature
9277786	Centrifugal Locks
9277790	"G" Weight
9277871	"G" Weight Lock
9278008	Housing
9278011	Slider Assembly (rear lock)
9317617	Set Back Pin Assembly
MS19060-4810	Ball Bearing

Also included are all associated springs, locks and closures for the above assembly.

Optional parts are:

Closure firing pin, Part No. 9317591
Rear locks, Part No. 9278057
Slider, lock, Part No. 9278060
Rear lock springs, Part No. 9278109
And all associated closures

Also included for the optional parts are all associated springs, locks and closures for the above assembly.

The test sample must not contain the staked disc closure, Part No. 9277812 for this test.

The slider shall be maintained in the correctly oriented position during the arming and non-arming tests. The contractor, when necessary, may utilize an interim slider lock for test purposes.

The S&A assembly shall be tested as follows using approved spin and testing equipment.

The S&A, after secured nesting in the spin fixture, shall be spun up to the arming speed specified on the applicable drawing. To verify that the set back pin is in the non-armed position, the contractor shall attempt to arm the "G" weight (Dwg. 9277790), by applying the minimum arming force as stated on the applicable drawing. The S&A sub-assembly shall not arm. The force on the "G" weight shall then be released.

The set back pin assembly (Dwg. 9317617) shall be moved to the armed position and held in place. Arming force shall be as specified in Drawing 9278015.

The S&A shall be spun to the non-arming speed as specified on the applicable drawing. The set back pin shall remain in the armed position. The contractor shall attempt to arm the "G" weight (Dwg. 9277790), in accordance with the minimum all-arm requirement as stated on the applicable drawing.

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While complying with the above, neither the "G" weight, Part No. 9277790, the ball bearing, Part No. MS-19060-4810, or the slider assembly, Part No. 9278011 shall move out of the safe state, i.e., into the armed position or in a position resting on the centrifugal locks. Duration of the time of spin for the test equipment shall be approved by the technical agency of 6.5.

After or during this spin test, the sub-assembly shall be examined for evidence of arming either in or out of the test nest.

Any S&A failing to comply with any or all of the non-arming requirements shall be classed defective and removed from the lot. (Non-destructive test)

4.5.2 Arming of safe and arming subassembly. The subassembly, consisting of the parts specified in 4.5.1 shall be spun up to the arming speed specified in Dwg. 9278015 using approved test equipment. When at the required arming speed and the set back pin assembly, Part No. 9317617, in the armed position, the "G" weight Part No. 9277790, shall be moved into the armed position and the slider assembly, Part No. 9278011, shall move into the full arm position. Failure of the sample to comply with all of the specified requirements shall cause the sample to be classified defective and removed from the lot.

NOTE: Arming forces on the "G" weight, Dwg. 9277790, and Setback Pin Assembly, Dwg. No. 9317617, need not be rigoriously controlled or measured during spin if it can be shown that the static force requirements are met.

The sample shall be disarmed at the end of the test by entering into further tests (See 4.5.3) or by other means as approved by the technical agency listed in 6.5. (Non-destructive test)

4.5.3 Non-arming of centrifugal locks (safe and arming sub-assembly.) An armed test sample, using approved test equipment shall be spun at the non-arming speed specified in (Dwg. 9278015). While at this speed, the slider assembly, Part No. 9278011, shall be moved into the non-armed position and the instrument performing this operation shall be withdrawn or removed. The slider assembly, without any physical restraint upon it shall not move into the full armed position but must remain in a non-armed position. This shall be indicated by the slider assembly coming to rest on top of the centrifugal locks Part No. 9277786.

The S&A shall be in full unarmed (safe) position after all tests have been completed. With the exception of the set back pin non-arming test specified in paragraph 4.5.1, (which may be performed during the arming sequence for convenience) all non-arming tests shall be performed after the arming tests. The order of the tests within the above arming and non-arming paragraphs is optional. Any S&A failing to comply with any or all of the requirements shall be classed defective and removed from the lot. (Non-destructive tests)

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4.5.4 Radiographic examination. The safe and arming assemblies radiographic plates shall be of a quality such that the Technical Agency of 6.5 or their appointed representatives shall be able to view all of the requirements to the Agency's satisfaction. The radiographic plates shall clearly define all explosive components; positively show the assembly is in the full non-armed condition; the flexible cable shall not be pinched, twisted or damaged; the MDF assembly shall not show evidence of any damage and all components shall be in the non-armed position. Any assembly failing to comply with all of the specified requirements shall be classified defective and removed from the lot. (Non-destructive test)

The vendor will have the option of substituting a paper base sensitive emulsion for the film base (Ref. MIL-STD-453) provided that this material produces the required definition specified above.

4.5.5 Resistance measurements.

4.5.5.1 Detonator continuity resistance. The measurement of the detonator resistance (value specified on Dwg. 9278015) shall be obtained on the final safe and arming assembly, (including staked shield assembly) utilizing an approved milliohmmeter whose excitation current does not exceed the no fire condition of the detonators. The resistance measurement shall be obtained across the eyelets on the flexible cable (Dwg. 9277774). Any assembly failing to comply with the specified resistance requirement shall be classed defective. The exposed conductor (shorting loop) in the flexible cable (Dwg. 9277774) must be cut to perform this test and soldered to reconnect the above conductor after completion of test. (Non-destructive test)

4.5.5.2 Detonator short circuit resistance. The soldered flexible cable assembly shall be measured for the specified resistance using an approved milliohmmeter whose excitation current does not exceed the no fire condition of detonators across the eyelet. This test must be performed after the device passes the conditions specified in 4.5.5.1.

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Any assembly failing to comply with the specified resistance requirements shall be classified defective. (Non-destructive test.)

4.5.6 In-line explosive functioning.

4.5.6.1 Explosive train. The armed safe and arming device shall be tested in an explosion proof test chamber. (To accomplish the test below, the soldered short on the flexible cable must be removed.) An eighty-three (83) microfarad plus or minus five percent (± 5%) capacitor charged to five plus or minus one tenth (5.0 ± 0.1) volts direct current shall be applied to the cable inputs which will cause the assembly to explosively function. The safe and arming device under test shall be inclosed by four (4) test plates located at the output of each of the four (4) main charge leads. Each test plate shall be hardened to a Rockwell Hardness of B80-B95 and be of a minimum thickness of six hundred and twenty-five thousandth's (0.625) inch. Further material references are contained in test method 301.1 of MIL-STD-331 except that the test plates shall be approximately 1-1/8 x 1-1/8 x 5/8 inch. As an alternate, a right circular cylinder of the above material measuring approximately 1-1/8 inch height (or an approved height adequate to contain the output), 5/8 inch minimum wall thickness and with an inside diameter of 1.170 + .010 inch may be used. The test shall be performed in accordance with the procedure specified in Method 301.1 of MIL-STD-331. In addition a visual examination shall be made of the chamber assembly to ascertain if all components have also functioned. Any sample failing to comply with all of the specified test requirements shall be classified defective. (Destructive test)

NOTE 1: The explosive train performance test may be performed on assemblies consisting of all explosive elements, housings, slider assemblies, slider locks and staked shield assembly assembled in the unarmed condition, and then armed remotely. The contractor shall describe he proposed method of remote arming (see para. 4.4.4). All explosive elements and metal parts shall be a representative sample drawn from each lot of parts that will consititute one lot of safe and arming assemblies (i.e.; complying with the lotting provisions of 4.4.1).

4.5.6.2 Aft ejection charge assembly. The aft ejection charge testing shall be performed in an approved chamber, as per figures 1 thru 6 for the required functioning and dent criteria. The test plate shall be in accordance with the requirements shown in the specification drawing, figure 4. Initiation of the aft ejection charge assembly shall be performed in an approved manner using a M100 electric detonator (Ref MIL-D-48120) or equivalent.

Any aft ejection charge assembly failing to comply with all of the specified requirements shall be classified defective. (Destructive test)

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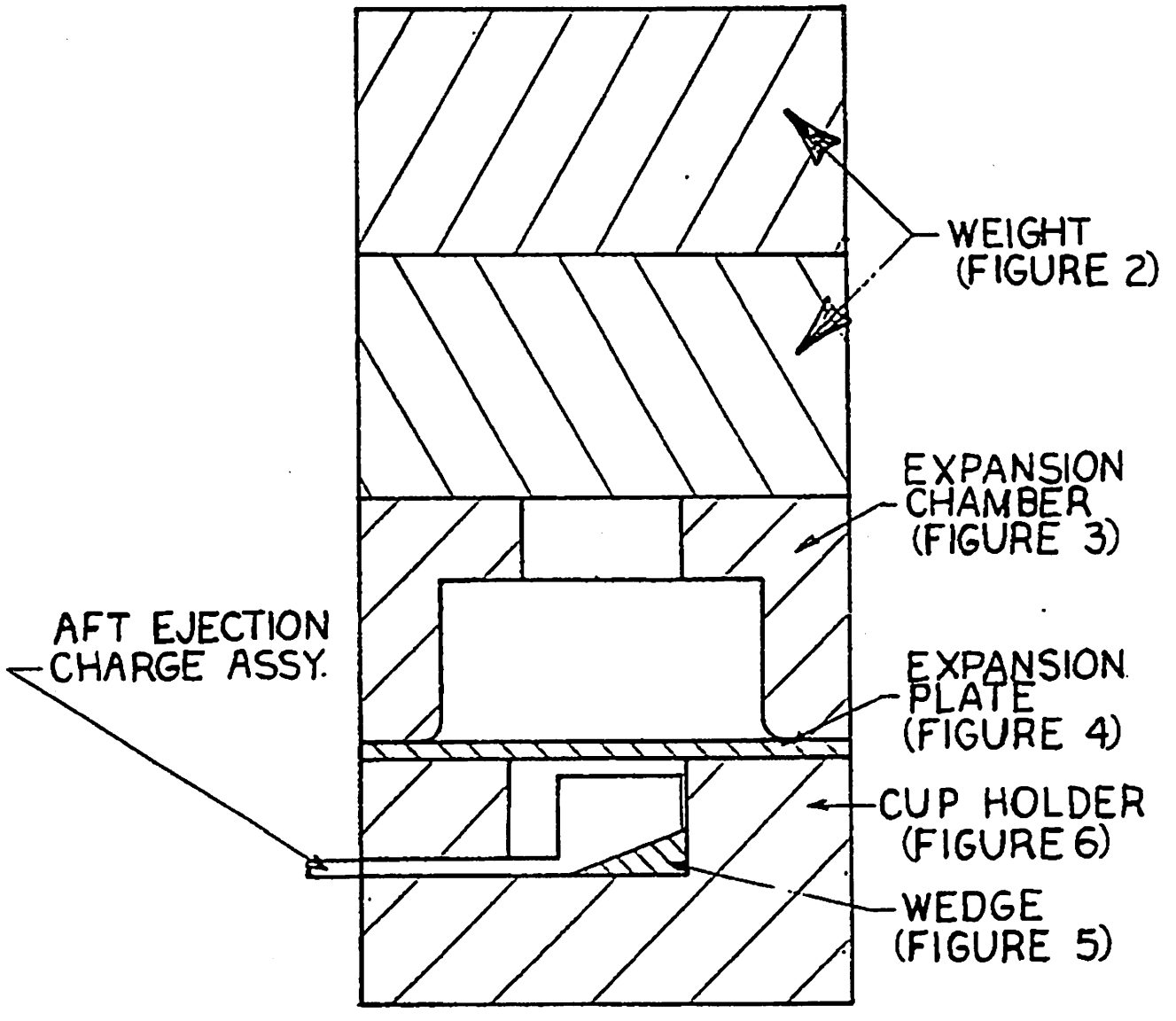
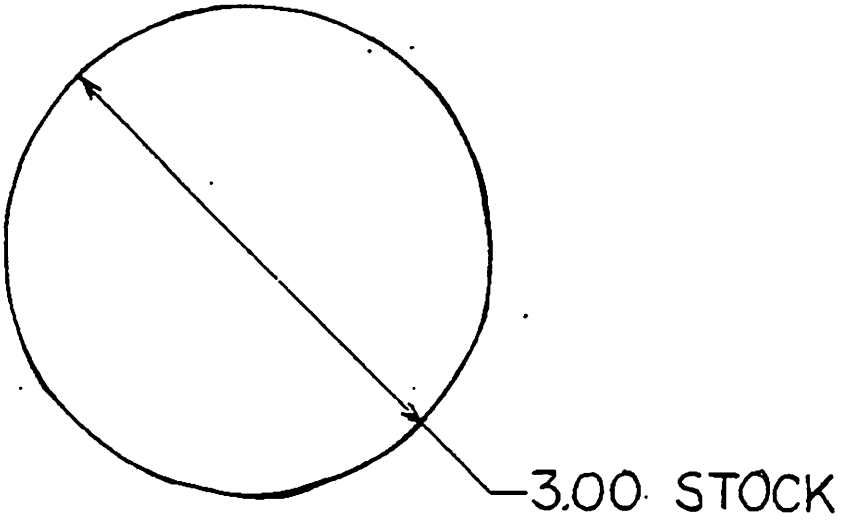
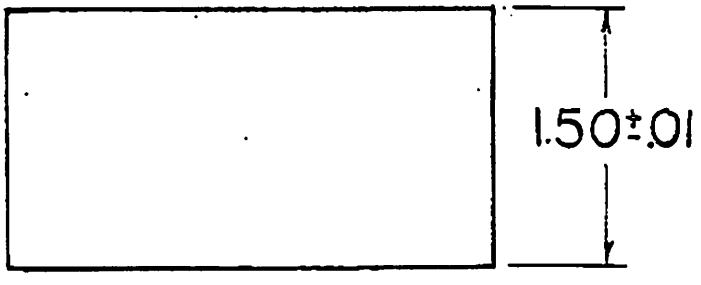


FIGURE 1

DENT TEST FIXTURE

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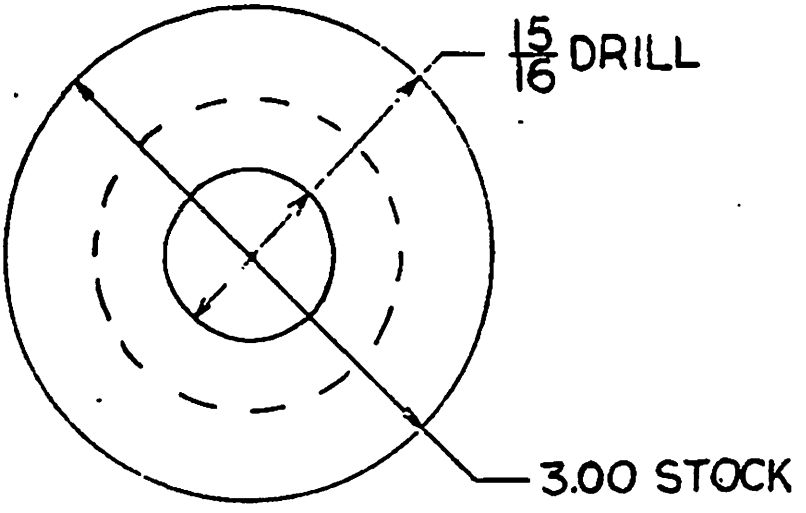


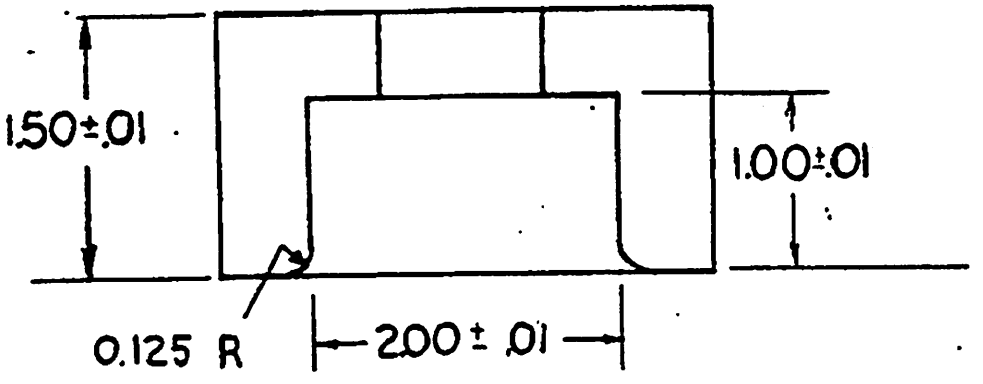
MATERIAL:
STEEL 1018
OR EQUIV.

WEIGHT

FIGURE 2

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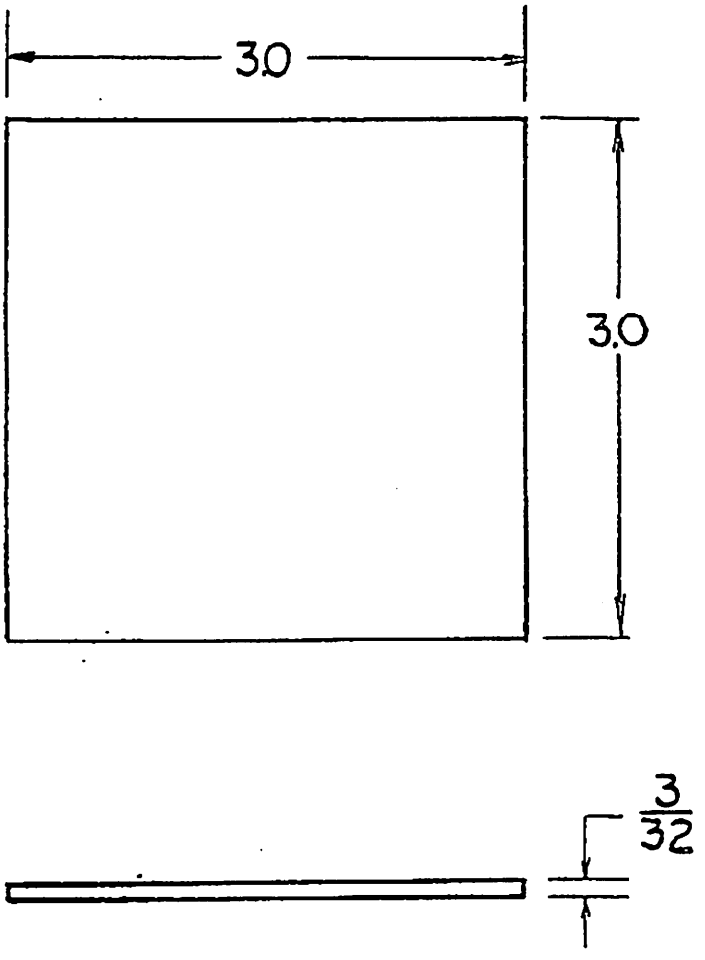




MATERIAL: 1018 STEEL
OR EQUIV.

EXPANSION CHAMBER
FIGURE 3

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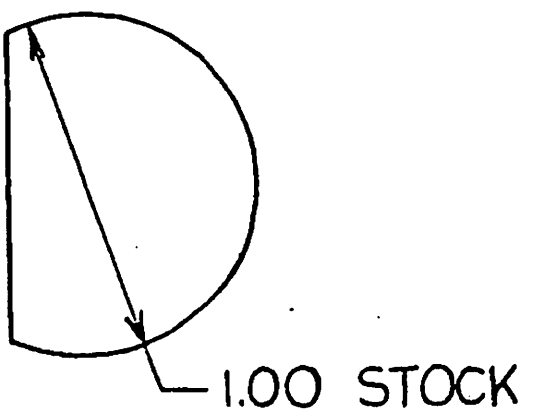


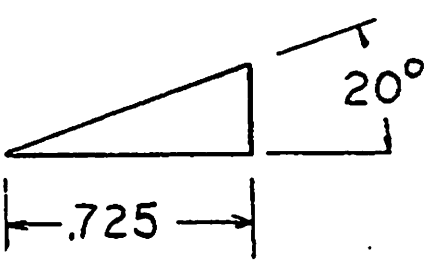
MATERIAL: ALUMINUM 6061-T6

EXPANSION PLATE

FIGURE 4

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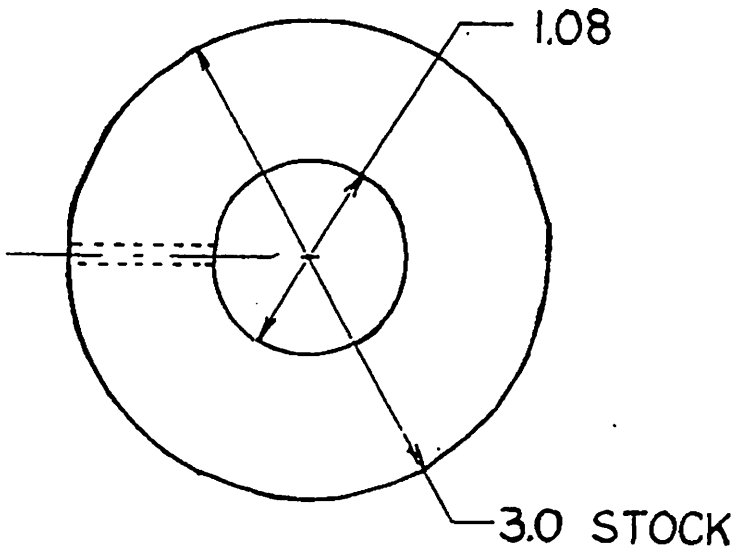


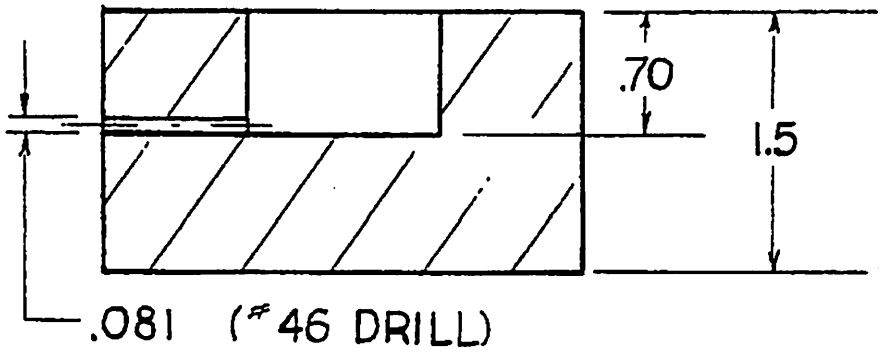


MATERIAL: 1018 STEEL
OR EQUIV.

WEDGE
FIGURE 5

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MATERIAL: 1018 STEEL
OR EQUIV.

CUP HOLDER

FIGURE 6

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4.5.7 Stake push off test. The test shall be performed using approved test equipment on sample of housings and closures staked in accordance with dwg. requirements, with no components within the housing. If any sample fails the push off test, all production following the last successful test shall be rejected.

4.5.8 Environmental and safety testing.

4.5.8.1 Jolt. The test shall be performed in accordance with the procedure specified in MIL-STD-331 Test No. 101.2. The safe and arming sample shall be classed defective if:

a. Any explosive lead detonates. (High or low order)

b. The slider assembly is in the fully armed position.

c. Both centrifugal locks do not mechanically overlap the slider.

d. The "G" weight and/or ball bearing are out of position.

For purposes of final safety determination, the sample shall be x-rayed for compliance for the above criteria.

NOTE: Care shall be exercised to assure no movement of parts shall occur during transportation to or during the x-ray operation. Samples shall not be returned to the lot after testing.

4.5.8.2 Jumble. The test shall be performed in accordance with the procedure specified in MIL-STD-331 Test No. 102.1. The safe and arming sample shall be classed defective if:

a. Any explosive lead detonates. (High or low order)

b. The slider assembly is in the fully armed position.

c. Both centrifugal locks do not mechanically overlap the slider.

d. The "G" weight and/or ball bearing are out of position.

For purpose of final safety determination, the sample shall be x-rayed for compliance for the above criteria.

NOTE: Care shall be exercised to assure no movement of parts shall occur during transportation to or during the x-ray operation. Samples shall not be returned to the lot after testing.

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4.5.8.3 Transportation vibration. The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 119. (Non-destructive test) Acceptance shall be as specified in MIL-STD-331, Test 119.

4.5.8.4 Static detonator safety. The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 115.1. The test fixtures as required by the test equipment paragraph of MIL-STD-331 shall be a closed cylindrical steel container. The inside diameter of the cylinder shall be 1.170 + 0.010 inches and shall completely cover the shield assembly. Units which have successfully passed the transportation vibration test in 4.5.8.3 may be used for this test. (Destructive test) Acceptance shall be as specified in MIL-STD-331, Test 115.1.

4.5.8.5 Test fixturing. The test samples, individually or collectively, shall be packed in a packaging material as used by the contractor to ship completed hardware to the load assembly facility for the test sequence of 4.5.8.1, 4.5.8.2 and 4.5.8.3. All parts and assemblies tested shall be scrapped and not returned to the lot.

4.5.9 Check test for deterioration of detonators and leads. (See MIL-L-63246, MIL-D-63197 and MIL-D-48120). If the total time between original acceptance of any detonator or lead lot and the assembly of the lot into the safe and arming device exceeds two years, or if the detonators or leads have been subjected to adverse conditions, however brief, at any time since previous test, the detonator or lead lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-L-63246 and/or MIL-D-63197, and/or MIL-D-48120. Immediately before the detonator or lead lot is assembled, the Government Inspector at the facility assembling the detonators or leads into the safe and arming device shall select the sample for deterioration tests and the contractor shall verify that the devices comply with the requirements of the above specifications.

4.5.10 Centrifugal lock hole inspection. Both centrifugal lock spring holders shall be removed using approved tooling and procedures. The housing shall be rotated and the centrifugal locks allowed to fall from the housing. A .2495 - .0005 inch pin gage shall fall freely (without applying pressure) to the bottom of the hole (shoulder). If one or both locks fail to fall out under their own weight or the pin gage fails to insert freely in one or both holes, the unit shall be classed defective and that quantity of subassemblies from which the sample was drawn shall be rejected. (Destructive Test)

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4.5.11 Ballistic tests. Safe and arming assemblies subjected to this test shall conform to drawing 9317620 and MIL-S-63383. All parts and explosive elements, with the exception of those unique to this configuration, shall be representative samples from each production lot. The safe and arming assemblies shall be forwarded to the load plant specified in the contract for assembly into M718 projectiles, part number 9317597, Projectile, 155MM, Test, Non-Hazardous, M718 Type. Testing of the projectiles and safe and arming devices shall be in accordance with MIL-P-63238, para 4.5.8, except that acceptability will be based on proper safe and arming assembly functioning determined after disassembly from the test mines.

4.5.12 Critical defect reporting. The contractor shall proceed as follows whenever a critical defect is discovered during the examinations or tests described in this specification:

a. Immediately notify the cognizant technical agency.

b. Tentatively reject and quarantine all previously accepted S&A subassemblies and assemblies in process.

c. Determine root cause of defect and proceed as directed by the contracting officer.

4.5.13 Solder workmanship. Examination for solder workmanship on the bottom side of the cable shall be performed on detonators and flexible cables and grounding clips soldered in accordance with the procedures normally followed during production with the detonators and grounding clips contained in re-usable housing (P/N 9278008) that contain no components and that have been modified by enlarging the detonator holes to facilitate removal of the detonators. The contractor shall recommend the extent of the housing modification and request approval in accordance with para. 6.3. Any unit failing to comply with the soldering requirements at the start of the shift shall result in the contractor not proceeding with production until corrective action is taken and a subsequent sample passes. Any sample failing during the subsequent examinations or at the end of the shift shall result in all production following the last successful examination being rejected. (Destructive examination)

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level - Commercial. The safe and arming device shall be preserved and packaged utilizing industrial procedures and in accordance with Code of Federal Regulations Title 49 100-199. These procedures shall be submitted and approved in accordance with Para. 6.5.

5.2 Packing.

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5.2.1 Level - Commercial. The safe and arming device preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with acceptable industrial procedures such as to prevent deterioration or damage during handling and shipment to the first destination. Industrial packing procedures shall be in accordance with Code of Federal Regulations Title 49 100-199.

5.3 Marking. Marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations Title 49 100-199.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The assembly covered by this specification is intended for use in the Projectile, 166mm, AT, M718/M741.

6.2 Ordering data. See MIL-STD-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Attn: DRDAR-QAT-I, Dover, New Jersey, 07801.

6.4 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonators and/or leads. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonators and/or lead or to the contractor assembling them into the S&A device. This shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the lot of detonators and/or leads was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations of tests to Commander, ARRADCOM, Attn: DRDAR-QAT-M/SA.

6.6 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, Attn: DRDAR-QAT-M/SA, Dover, New Jersey, 07801.

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6.7 Drawings. Drawings listed in Section 2 of this Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A228)

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